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Validity and objectivity: Reflections on the role and nature of Rasch models

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This paper discusses a number of methodological issues related to item analysis by Rasch models including the relationships between objectivity as defined by Rasch and the different ways construct validity is defined in psychometrics. To reconcile the different definitions of validity, the IRT models have to be extended to include exogenous variables in addition to items. One such family of extended models referred to as graphical IRT models is described. Graphical Rasch models are special cases of a graphical IRT model characterized by both objectivity and construct validity. Finally, a family of models called graphical loglinear Rasch models is described. Measurement by items from such models is neither valid nor objective according to formal psychometric definitions. The paper argues, however, that measurement is essentially valid and objective and that it is preferable to retain all items to preserve reliability rather than eliminate items to obtain measurements satisfying all the requirements of validity and objectivity.
Keywords: Construct validity, specific objectivity, nomological networks, Rasch models, graphical models, graphical Rasch models, graphical loglinear Rasch models.

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> *"A sensational innovation by Dr. Rasch … This ground-breaking work will be studied and used, not only in Denmark, but all over the world"* (Jolander, 1957).

> *"These four chapters [of Rasch(1960/1980)] describe the dawn of a new era in psychometric theory and practice. … They embody the essential principles of measurement itself, the principles on which objectivity and reproducibility, indeed all scientific knowledge, are based"* (Wright, 1980)

Introduction

The Rasch model has had a profound effect on psychometrics, item response theory (IRT) and scale validation since its publication in Rasch (1960/80). It is regarded by many as the gold standard against which summated scales summarizing item responses have to be assessed. The two quotations that preface this paper reflect sentiments and enthusiasm characteristic of the development initiated not only by the work of Prof. Rasch, but also by Rasch's charismatic personality. The quote from Jolander's (1957) report from a meeting of the council of the Danish Institute for Educational Research in 1957 is uncannily prophetic. Even today, one rarely finds an issue of journals like Psychometrika and Applied Psychological Measurement without at least one paper dealing with some aspect of Rasch modelling, and there are at least two journals[1] dedicated to the theory of Rasch models.

Item analysis by Rasch models serves two different purposes. The first is to provide calibrating equations relating total scores on educational or psychological tests to estimates of the value of the latent trait variable underlying the responses. These procedures can be characterized as person measurement procedures. For this purpose, item analysis also estimates item parameters either prior to estimation of the latent traits or jointly with the latent trait estimates. Estimation of item parameters is, in most cases, the means to the ends. They are rarely of interest in themselves.

The second purpose of item analysis by Rasch models is scale validation. Items in Rasch models are characterized by a number of very attractive properties. Measurement by Rasch models is not only construct valid and objective, but the Rasch model is also the simplest of all item response models in both mathematical and statistical terms. And it requires no assumptions concerning the distribution of the latent trait, whereas most other item response and factor analysis models require that the latent trait is normally distributed. The second purpose of the analysis is to make sure that items possess these properties by an examination of the fit of item responses to the model. Most procedures for analysis of fit also use estimates of item parameters, and some use person parameter estimates.

The Rasch model is usually considered to date back to 1960, the year when Rasch's book on probabilistic models for some intelligence and attainment tests was published. However, the work on the book and the development of the models goes much farther back. The first known written example of the model, dating back to 1952 or 1953, is shown in Figure 1. A technical report describing the model was published in Danish in 1958 (Rasch,1958).

Independent of the development of Rasch models, many other important advances with a lasting effect on psychometrics and scale validation were pioneered at the same time. The paper on Cronbach's Alpha, one of the preferred measures of reliability, was published in 1951 (Cronbach, 1951/2006). A seminal paper on construct validity was published by Cronbach and Meehl in 1955 (Cronbach and Meehl, 1955/2006). Latent structure models, forerunners of confirmatory factor analysis and item response models, were also developed during the 50'es (Lazarsfeld 1950, 1959). The Rasch model shares many of the assumptions underlying the notions of both construct validity and latent unobservable *variables*, but at the time of its conception it was distinct from these developm-ents. Rasch rejected reliability as a measure of interest[2] in connection with scale development because reliability as defined above does not describe the test as such, but rather the way the test functions in a specific population. Reliability, in Rasch's understanding, was not an objective statement about the test. We can also assume that he was not concerned about validity as it was defined in the fifties, since there is no reference to this concept at all in his writings. And finally, the person parameter in the Rasch model in Rasch (1960/1980) is a fixed parameter without a distribution. It is not the outcome of a latent *variable*. The symmetry between the item parameter and the person parameter was, to Rasch, a very important point (Rasch, 1977/2006)

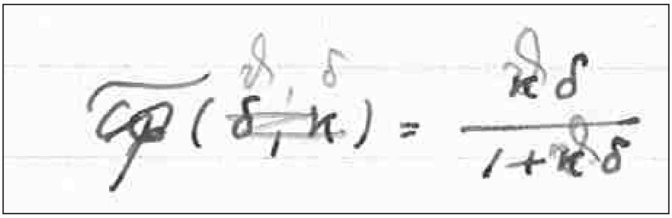

Figure 1. The Rasch model anno 1952/53

Rasch models and/or validity?

Today, the Rasch model is completely embraced by modern psychometrics. For this reason, there appears to be some confusion concerning the role of the Rasch models during item analysis. A common misunderstanding is that the main purpose of the Rasch analysis is to check that the total score is a valid and uni-dimensional measure of the latent trait. While this is not completely incorrect, a closer look at the notion of validity will immediately show that there are several validity issues that cannot be addressed by a Rasch analysis, and that validity does, in fact, not require that items fit a Rasch model at all. The answer to the question about what we get out of a Rasch analysis, when validity does not insist on a Rasch model, is that measurement by Rasch items in addition to validity also has two useful and fundamental properties: objectivity and sufficiency.

Measurement is specifically objective in the sense that statements concerning persons and items do not depend in a systematic way on arbitrary choices made by the person who developed the test or the person using the test as long as you stay within the specific frame of reference in which the test was developed. Measurement of depression may be valid and objective as long as you only try to measure depression among persons who are suffering from depression. If a depression test is used among persons who do not suffer from depression at all, there is no guarantee of objectivity: a higher score on the depression test for one such person compared to another person can not mean that the first person is more depressed than the other since neither person suffers from depression.

It is an underlying assumption behind most psychological tests that the score provides important information on the trait, property or syndrome that the test is supposed to measure. A high score on the MH scale measuring mental health discussed later in this paper means that the person has good mental health. The MH score is sufficient if additional information on the item responses lying behind the score does not add anything to the assessment of the level of mental health. Consider, for instance two persons. The first person is nervous all the time and never calm and peaceful, but also happy all the time, never down in the dumps, and never downhearted and low. The second person is sometimes nervous, sometimes down in the dumps, sometimes calm and peaceful, sometimes downhearted and low and sometimes happy. Both persons have MH scores equal to 15. The two persons are therefore at the same level of mental health. If the differences in the responses to the separate items tells us nothing that we can use to distinguish between the mental health of the two persons then the MH score is sufficient.

Measurement by Rasch items is in some sense ideal measurement because measurement is valid, objective and sufficient. A closer look at these properties will reveal, however, that objectivity and sufficiency can be attained in cases where items are not valid according to psychometric definitions. This raises yet another question: to which degree are the psychometric definitions of validity fundamental to measurement? Some of the requirements of construct validity are methodologically convenient because the statistical analysis becomes much easier if the requirements are satisfied. The question is whether they are more than assumptions of convenience. Are they really indispensable? Is it impossible to use item responses for measurement if some of the conventional assumptions of validity are violated? The main intention of this paper is to throw some light on this question. In order to do so, we will look at a family of item response models referred to as graphical loglinear Rasch models (GLLRM) which we claim provide essentially valid and essentially objective measurement even though some of the fundamental assumptions of item response models do not hold.

The example

We use the results from an analysis of data on one of the subscales of the short-form 36-item General health Survey questionnaire (SF-36), a widely used instrument measuring different aspects of subjective health status (Ware & Sherbourne, 1992). The subscale summarizes responses to the following five questions:

How much time during the past month
- Have you been a very nervous person? (NERVOUS)
- Have you felt so down in the dumps that nothing could cheer you up? (FELTDOWN)
- Have you felt calm and peaceful? (CALM)
- Have you felt downhearted and low? (DOWNHEAR)
- Have you been a happy person? (HAPPY)

The six ordinal response categories ranging from "All of the time" to "None of the time" were scored from 0 to 5 for responses to the first, second and fourth question, and from 5 to 0 for the third and fifth question. The total score summarizing item scores for the five questions is used as a measure of mental health with a high score indicating good mental health.

The data used to validate the MH scale originated in a health survey in Denmark in 1995. A total of 2525 persons responded to all five questions. The data was analyzed by graphical loglinear Rasch models (Kreiner & Christensen, 2002, 2004 and 2006). The interpretation of these models will be discussed throughout this paper. The final GLLRM is in itself of considerable interest. The total score summarizing responses from this model is statistically sufficient, but it is neither valid according to formal psychometric criteria, nor objective according to Rasch's definition of this concept. Despite this, we will argue that measurement by the MH scale is both essentially valid and essentially objective.

Even though we are only using the SF-36 questionnaire to discuss what graphical loglinear models tell about measurement, the results of the item analysis may also be of interest to readers familiar with SF-36. Some of the results are therefore included in Appendix C.

Construct validity

Construct validity is a difficult and challenging concept. In a recent paper on the integration of psychology and psychometrics, Borsboom (2006) describes construct validity as a "black hole from which nothing can escape: Once a question gets labelled as a problem of construct validity, its difficulty is considered superhuman and its solution beyond a mortal's ken". A quick look at papers describing analyses of construct validity will show that this appears to be true. There are many different ways of thinking about construct validity. We will here entertain but two different definitions: one which focus on associations with other variables (Cronbach & Meehl, 1955/2006) and another addressing the theoretical construct underlying the responses to items (Rosenbaum, 1989).

Both points of view are based on the assumption that the construct being measured has a sound theoretical foundation, and both points of view consider scale validation as a confirmatory analysis. Confirmatory factor analysis, for instance, is regarded as an analysis of validity, whereas exploratory factor analysis is not. The first point of view is based on ideas from classical test theory and Cronbach and Meehl's definition of a nomological network summarizing known facts about the relationship between the theoretical construct and other variables. Congeneric test theory[3] defines the second viewpoint. From this perspective, construct validity is a question of what item responses depend on; measurement is construct valid if item responses depend on nothing but the construct, and invalid if other variables have systematic effects on responses.

The relationship between the different notions of validity has perhaps not always been crystal clear, and researchers, in most cases adhere to one, but rarely to both, definitions of construct validity. One of the reasons for this is probably that historically, classical test theory, confirmatory factor analyses and item response analyses have been regarded as competing and somewhat incommensurable approaches to scale validation. Today, these antagonists have to a large degree been reconciled.

Theories

Constructs do not exist in a void. They are assumed to be components in psychological theories together with other variables, including background variables with an effect on the distribution of the construct, concurrent variables associated in a non-causal way with the construct, and posterior variables depending on the construct. The items that are supposed to provide indirect measurement of the unobservable latent construct are also part of this framework. It may be true, as claimed by Borsboom (2006), that there is very little evidence of elaborate psychological theories underlying development of psychological tests. This does not mean that the scales are theoretically unfounded, but theories are in many cases imprecise low-level theories based on common sense arguments. For this reason, scale validation is in many cases as concerned with the validity of the construct as with the validity of the scale.

One of the many things pointed out by Borsboom is the fact that congeneric test theories, item response theory and factor analysis presuppose theories that to some extent are causal theories in which the constructs are the causes of item responses.

Another important component of the theoretical framework is a theory of response behaviour. This component is at best implicit, but in most cases not at all considered in applications of item response and factor analysis models. Constructs may, for instance, be dispositions where item responses do not exist, unless they are provoked in some sense. The response to an item in a math test does not exist before the item has been written and presented to the student. The response to a question in an attitude scale may not exist unless provoked. Items may, however, also refer to characteristics and properties of persons that are assumed to depend causally on the construct. Nervousness, for instance, is both an item in the mental health scale and a state with duration. Nervousness exists before the question is asked and persists for some time after the answer has been given. The difference between these two types of item responses is particularly important in connection with longitudinal studies. In a longitudinal study of cognitive function, one may perhaps assume that the solution to an item in a cognitive test has been forgotten when the test is retaken a year or two later. The response to the item will therefore only depend on the current cognitive state. In a longitudinal study of mental health, the second response to the question of nervousness will depend both on the current mental health state and on the response the first time the question about nervousness is asked.

A fundamental assumption behind all types of item responses is that behaviour is rational. We assume that students do their best to solve the problems in educational tests. Likewise we assume that persons responding to the MH questions try to do so in a meaningful and consistent way. The way the questions are presented must therefore influence response behaviour. If questions are asked during an interview, where persons cannot go back and change their responses in the light of responses to other questions and where they may have problems remembering exactly what was asked beforehand, responses to different questions may to some degree be assumed to be independent. If items are presented together in a postal questionnaire, where persons can compare the different questions against each other, we must assume that they try to respond in a way which is logically consistent. It is difficult to see how one could claim that he or she has been "downhearted and low" all of the time, but "down in the dumps" none of the time. It is therefore no surprise that the analysis of the MH scale did, in fact, disclose evidence of local dependence. Given the contents of these items, we prefer to interpret this as result of response dependency rather than as evidence of an additional latent trait.

Having recognized the causal relationship between the construct and the item responses, it is not difficult to accept that the theoretical context itself may be partly causal, distinguishing between prior variables that may have an effect on the construct, concurrent variables that may be associated with the construct in a non-causal way, and posterior variables that may depend on, but cannot have an effect on, the construct. This leads to the definition of the nomological network.

The nomological network

Cronbach and Meehl (1955/2006) stress that no theoretical construct is defined without a context and that the meaning of the construct, to a very large degree, is defined by the variables which are known to be associated with the construct. They refer to the network of variables known to be associated with the construct as the *nomological* network of the construct and argue that construct validity requires that the measure of the construct has to be associated with all the variables known to be connected to the construct in the nomological network. In other words, analysis of construct validity is nothing but a conventional, although extended, analysis of criterion validity covering convergent validity, discriminating validity, predictive validity and "known groups" validity.

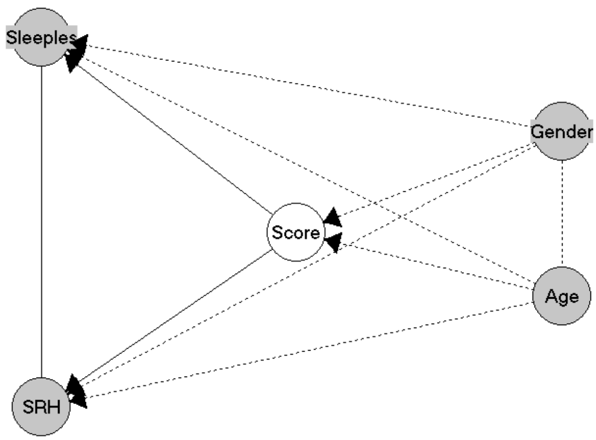

Figure 2. A nomological network for the MH subscore. Mental health is assumed to have an effect on sleeplessness (Sleepless) and Self reported Health (SRH), and may depend on Age and Gender. Dotted lines indicate associations that are not required according to the nomological network

To talk about a theory of mental health in connection with the SF-36 subscore is perhaps farfetched, but the SF-36 obviously does not exist in a void, and common sense considerations may at least serve as a sketch of a theory. Figure 2 shows a reduced version of such a network and may serve to illustrate the idea of the network. The network contains two prior variables that may be, but do not have to be, related to mental health. It also contains two posterior variables, Self reported Health (SRH) and sleeplessness, which common sense tells us are monotonously related to mental health; the worse the mental health, the worse the self reported health and the problems with sleeplessness. In this network, construct validity is reduced to a problem of predictive validity: The MH score is not construct valid if it is not statistically related to both posterior variables.

The nomological network is usually regarded as nothing but a visual illustration of the theoretical context. Networks like the nomological networks are in other contexts used to define joint statistical models for all the variables in a network. The best known example of such models is the family of structural equation models (Bollen, 1989). Another example is the family of graphical models (Lauritzen, 1996) discussed below. If the nomological network is supposed to correspond to a graphical model, then correlation itself is not conclusive evidence of construct validity. In a graphical model, it is required that the correlation between the score and a variable associated with the construct can not be explained by the fact that the items and the score are correlated to other variables. The *conditional* or *partial* correlation between the score and the variable must not be equal to zero. The nomological network in Figure 2 thus insists that the partial correlation between the MH score and the self reported health, controlling for sleeplessness, age and gender, must be positive.

Local independence and no differential item functioning

A very different approach to construct is adopted by Rosenbaum (1989). According to Rosenbaum, measurement is construct valid if, and only if, the unobservable construct is the *only* systematic factor influencing responses. Rosenbaum's criteria can be summarized as follows:

1) *Unidimensionality*. There is a single unobservable *latent* variable lying behind the item performance on the test. This variable, often referred to as Theta (θ), represents the construct the items are meant to measure. All conventional psychometric models assume that θ is measured on an interval scale with an arbitrary origin and an arbitrary unit, so that differences between persons are given by the differences between θ values.
2) *Monotonicity*. Item responses are positively related to the latent variable.
3) *Local independence*. Items are conditionally independent given the latent variable.
4) *No DIF*. The items are conditionally independent of other variables given the latent variable.

Borsboom et. al (2004) and Borsboom (2005) define validity in a way which is very close to Rosenbaum's definition. Borsboom emphasises the assumption of a causal relationship between θ and the items and also calls attention to the necessity of including considerations of response behaviour in the theoretical context of the measurement instrument.

The concept of *conditional* independence is central to the understanding of Rosenbaum's definition. Conditional independence means that the association between A and B can be fully explained by C. The assumption of local independence is therefore an assumption that the existence of θ explains why items are correlated. The assumption of no DIF is the assumption that θ provides the full explanation of why items and exogenous variables are correlated. If item responses depend on age, for instance, it should be because the latent variable depends on age.

The monotonicity assumption, 2), serves two different purposes. First, monotonicity means that the *expected* total score is an increasing function, F(θ). For a given observed score, s, there exists one θ value for which F(θ) = s that can be used as the measure of θ. Seen from this point of view, monotonicity is an assumption of convenience. Measurement – estimation of θ – may also be possible in situations where monotonicity does not apply, but the estimation procedure has to be based on the complete item response profile, rather than on the total score. Second, monotonicity connects Rosenbaum's definition of construct validity to construct validity defined in terms of the nomological network: if monotonicity applies, and if a variable X is known to be monotonously related to θ, then X has to be monotonously related to the total score. In other words, according to Rosenbaum, assumptions about monotonous associations between the construct and other variables plus construct validity define the nomological network

The assumption of no DIF, 4), is required for measurement to be unconfounded by other variables. The no DIF assumption is strongly related to the requirement of objectivity that is important for the theory of Rasch models defined below. If there is DIF, then measurement will not be objective.

In recent years, there has been a strong development of statistical models defined by assumptions of conditional independence. These models are called graphical models, because they are characterized by networks where the nodes of the network correspond to variables and where a missing link between two nodes means that the two variables are conditionally independent given all the other variables in the network. Networks defining graphical models are often referred to as independence graphs or Markov graphs. The nomological network in Figure 2 could be a Markov graph, if the theory of graphical models (Lauritzen, 1996) did not happen to be more recent than the notion of the nomological network. The nomological network of Cronbach and Meehl is nothing more than a picture of the associations among variables surrounding the construct. The Markov graph, on the other hand, defines a complete multivariate statistical model.

Since conditional independence is also central to Rosenbaum's definition of construct validity, it follows that graphical models may be regarded as natural frameworks for psychometric statistical models defined by these assumptions. A Markov graph encapsulating Rosenbaum's definition of construct validity for the MH subscale is shown in Figure 3. The graph contains one latent variable, θ, the items and two exogenous variables, Age and Gender. Items are connected to θ, but disconnected from each other and disconnected from Age and Gender. The graphical model defined by Figure 3 for the MH items therefore meets requirements 1), 3) and 4) of Rosenbaum's definition of construct validity.

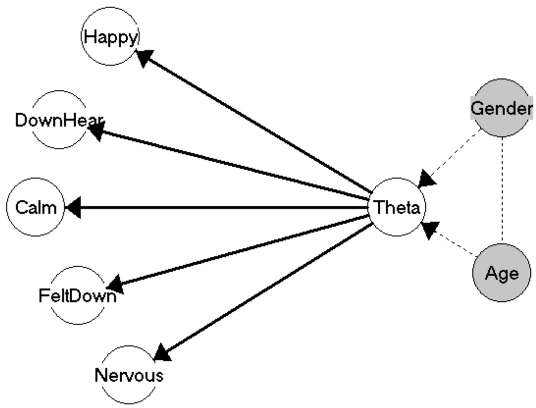

Figure 3. The IRT graph of the MH scale. A graphical model for a unidimensional scale with locally independent items without DIF

We refer to a graph defined by unidimensionality, local independence and no differential functioning as an IRT graph. Note that the "Sleepless" and SRH are not included in the IRT graph since they are causally posterior relative to the items and therefore cannot confound measurement.

In addition to defining the graphical model shown in Figure 3, assumptions 1) - 4) together with the assumption of causality tell us that the relationship between the items and the latent trait can be described by a collection of independent regression models. The first describing the way the first item depends on the latent variable, the second describing how the second item depends on the latent variable, and so on. If items are dichotomous, with 0 indicating a negative and 1 a positive response, then the regression models describe the probabilities, $P(Y_i = 1|\ \theta)$ where Y_i is the i'th item and θ is the latent variable. Assumption 2) that item responses are monotonously related to θ only means that these probabilities must be increasing functions of θ but makes no assumption about the actual type of function. Models without such assumptions are referred to as "Mokken models" or monotonous nonparametric IRT models (Sijtsma and Molenaar, 2002). Strictly speaking, construct validity therefore only requires Mokken models.

Measurement in Mokken models is always in terms of the total score because there is no way to estimate the value of θ. If an estimate of θ is required, it is necessary to use a parametric item response model based on a specific mathematical function. The natural first choice of a parametric IRT model for dichotomous items is a model describing the relationship between items and θ by logistic regression models

$$P(Y_i = 1 \mid \theta) = \frac{\exp(\alpha_i + \beta_i\theta)}{1 + \exp(\alpha_i + \beta_i\theta)} \qquad (1)$$

The model defined by (1) satisfies the monotonicity requirement. In item response theory this model is referred to as the two parameter model because it contains two parameters for each item and usually rewritten as

$$P(Y_i = 1 \mid \theta) = \frac{\exp(\beta_i(\theta - \alpha_i))}{1 + \exp(\beta_i(\theta - \alpha_i))} \qquad (2)$$

where the effect parameter, β_i, is usually called the item discrimination, and the item parameter, α_i, is referred to as a threshold parameter corresponding to the θ value for which the probability of a positive response is exactly 50 %.

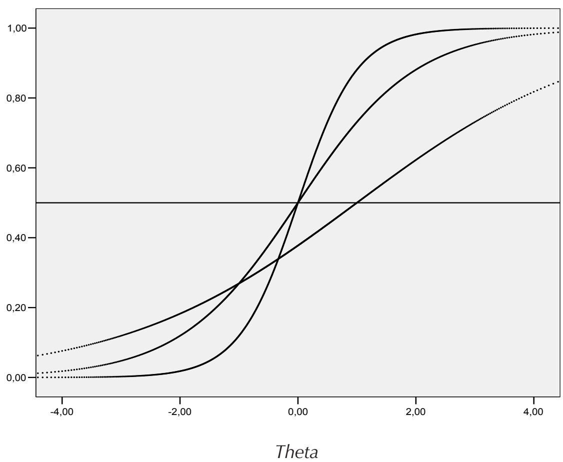

Figure 4. Item characteristic curves for three items with item discrimination equal to 0.5, 1 and 2.

A plot of the probability of a positive response to an item against the value of the latent variable is usually referred to as an item characteristic curve. Figure 4 shows three item characteristic curves for three items with different item discriminating powers. The probability of a negative response to an item at a given θ level may be regarded as a measure of the difficulty of the item. Model (2) not only defines models where the difficulties of items depend on the latent traits. In Figure 4, the item with the strongest item discrimination is the most difficult item among persons with very low θ-values but the easiest item for persons with very high θ-values. In fact, if all items have different item discriminations, the rank order of items according to difficulty would be exactly the opposite for persons with very low θ-values as for persons with very high θ-values. This aspect of the two-parameter IRT model obviously makes no sense in most cases. To avoid this problem we have to assume that the β parameters of (2) are the same for all items. This leads to the model referred to as the Rasch model described below. From this point of view we may characterize the Rasch model as the only parametric IRT model where the idea of item difficulties makes sense.

Graphical IRT models – no differential item effect

A graphical IRT model combines the nomological network of Cronbach and Meehl with the IRT graph defined by Rosenbaum's definition of construct validity. The model decomposes into a measurement component describing the effect of the latent variable on the items and a structural component corresponding to the nomological network. The model is similar to and shares some properties of structural equation models. Graphical IRT models are, nevertheless, different from structural models and may be considered a more natural framework for construct validation because they refer explicitly to conditional independence and generalize painlessly to extended graphical models for Rasch models where there is an explicit reference to the total score.

The IRT graph of the complete nomological network is shown in Figure 5. The difference between the IRT graph in Figure 5 and the IRT graph in Figure 3 is that the two posterior variables, SRH and sleeplessness are included in this model.

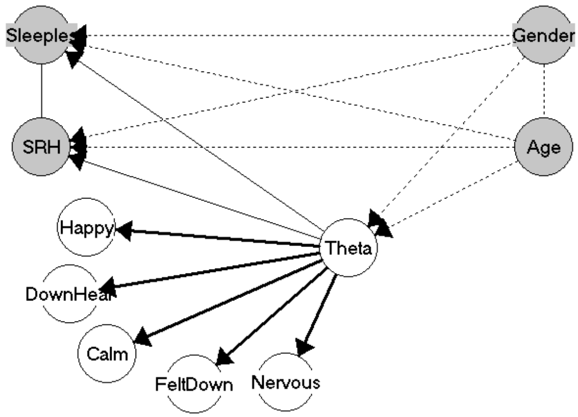

Figure 5. The IRT graph representing the complete nomological network of the MH scale

Graphical Rasch models

The conventional Rasch model for dichotomous items is a model where the probability of positive responses corresponds to (2), except that all items discriminate in the same way. Under this assumption, we may eliminate the common item discrimination parameter by a rescaling of the latent variable as shown in Appendix B.

The differences between the nonparametric IRT models defined by the IRT graphs in Figures 3 and 5 and the Rasch model boil down to two fundamental properties

5) Measurement by the Rasch model is specifically objective
6) The total score is statistically sufficient

To Rasch, the requirement of objectivity meant that measurement – estimation of θ – must not depend systematically on the choice of items nor on the distribution and sampling of persons. Objectivity in this sense means that measurement does not depend on actions reflecting arbitrary choices made by the user of the instrument, as long as the choice of items and sampling of persons are made within the specific frame of reference that the theory of the construct belongs to.

Statistical sufficiency means that the conditional distribution of the set of item responses given the total score does not depend on the latent variable. There is therefore no additional information on the value of θ to be obtained from response *profile* in addition to the information provided by the total score. If the total score is sufficient it follows, therefore, that the measure of θ is a function of the total score and that the item response profile can be disregarded.

The fact that assumptions 5) and 6) not only follow from the Rasch model, but that the Rasch model follows from either objectivity (Rasch, 1977/2006) or sufficiency (Andersen, 1977) is one of the fundamental results from the theory of Rasch models. The proof of these results, however, presupposes unidimensionality, monotonicity and local independence. The correct way to state the results would therefore be the as follows:

(i) Items are construct valid according to Rosenbaum (1989) and measurement is specifically objective if, and only if, item responses fit a Rasch model.
(ii) Items are construct valid according to Rosenbaum (1989) and the total score is statistically sufficient for θ if, and only if, item responses fit a Rasch model.

The assumption of no DIF may be regarded as implied by the assumption of objectivity. Note, however, that the conventional Rasch model only contains a latent trait variable and items. The conventional Rasch model describes measurement in a vacuum, disconnected from the rest of the world. Questions concerning the nomological network and questions concerning DIF cannot be formulated and formalized within the frame of inference defined by Rasch models.

For that to be possible, we need models which include explicit references to exogenous variables. Figures 3 and 5 define such extensions of the general IRT models. To the models defined in these graphs, we add the extra assumption that the construct has the same effect on all items. The result is what we call a graphical Rasch model. The model is defined by the same assumptions as the models in Figures 3 and 5 with the added assumption that the score is statistically sufficient. This score is added as a variable in Figure 6. We can show (Kreiner and Christensen, 2002) that the assumption of sufficiency means that the items are conditionally independent of θ and all other variables. For this reason, the score has to separate items from θ and all the other variables in the graph.

It follows from (ii) that the Rasch model is characterized by sufficiency *and* construct validity. That is to say, the graphical Rasch model is a graphical model characterized by two graphs; the IRT graph of Figure 5 representing construct validity and the Markov graph of Figure 6 representing sufficiency. We refer to the second graph as a Rasch graph, because we regard the sufficiency of the score as the fundamental property of the Rasch models.

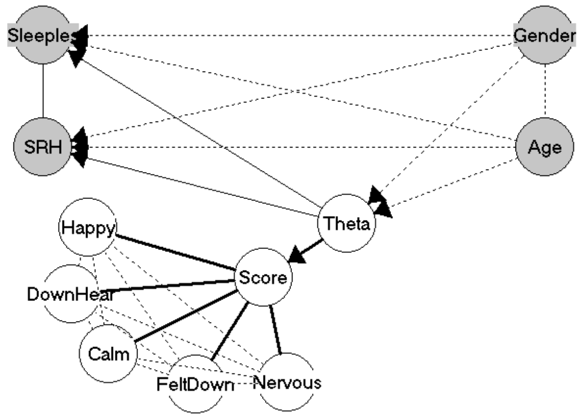

Figure 6. The Rasch graph. The sufficient score separates items from all the other variables

The (graphical) Rasch model for dichotomous items has been extended in several ways to models for polytomous items with more than two response categories. The Partial credit model (PCM) is a unidimensional Rasch model for ordinal item responses. Some details of the model may be found in Appendix B. The "Partial credit" terminology was coined by Masters (1982) who showed how one type of response behaviour might lead to the probabilities of the PCM model. Other types of response behaviour may, however, lead to the same model. Therefore, the PCM model does not presuppose a specific type of response behaviour.

Scale validation

Scale validation by item analysis is a confirmatory statistical analysis attempting to check the assumptions expressed in Figures 4 and 6. The analysis is naturally divided into two parts. First, analysis of criterion validity checks whether correlations predicted by the graph actually exist. Second, the item analysis, checks that there is no evidence against the independencies predicted by the graph.

The technical details of the analysis will not be discussed in this paper[4]. Note that the item analysis by Rasch models is typically more elaborate and comprehensive than item analysis by other types of IRT models because the Rasch model has a wider range of overall test-of-fit statistics, item fit statistics sensitive to DIF and differential item discrimination, and tests of local independence (Fisher and Molenaar, 1995, Smith, 2004 and Kreiner and Christensen, 2004). The wide range of statistical methods for the Rasch model is one of the reasons why Rasch models are often used for construct validation, even though, strictly speaking, construct validation does not require Rasch models.

The analysis showed that the MH score correlates with both SRH and "Sleepless". The MH is therefore criterion valid. The Rasch model was rejected due to strong evidence of local dependency and some evidence of DIF relative to both Age and Sex. From a strictly psychometric view, the MH is not a construct valid scale.

Graphical loglinear Rasch models

The Rasch model is the one and only model which is characterized by construct validity and either sufficiency or objectivity whereas other conventional IRT models provide construct valid measurement which is neither objective nor provided by a sufficient score. This raises a new question. Is it conceivable that measurement could be sufficient and/or objective without being construct valid? The answer to this question is confirmatory. There exists a family of item response models referred to as graphical loglinear Rasch models (GLLRM) where the total score is sufficient as in the conventional Rasch model. Measurement by items from a GLLRM is, however, neither objective according to Rasch's definition of the term nor construct valid according to Rosenbaum because items may both function differentially and be locally dependent. The requirement of these models is that both DIF and local dependence is uniform. Uniform DIF and uniform local dependence and their effect on validity and objectivity will be informally defined in this section. The mathematically inclined reader may find the formal definitions in Appendix B.

Uniform DIF

Construct validity requires that the probabilities of the different responses to an item depend on nothing but the latent variable. If it turns out, that the probabilities also depend on the gender of the respondent we say that the item functions differentially relative to sex. Differential item functioning may be either uniform or non-uniform. DIF is uniform relative to sex, if the Rasch model fits the responses to item for men and women separately, such that the DIF effect simply means that the *difficulty* of the item depends on sex in such a way that the effect of sex on the difficulty is the same for all levels of the latent variable. If there are no other problems with the items, and if the study in which item responses were observed only included men, there would consequently be no evidence against construct validity. In this sense, variables causing uniform DIF effects may be regarded as variables defining restricted frames of reference in which measurements appear to be valid. Measurement is valid within the different frames of references, but comparisons of measurement across frames of references will be confounded by the DIF effect.

Uniform LD

Construct validity requires also that the latent variable explains all the correlation between items. If this is not the case, we say that items are locally dependent. If the correlation between the two items is the same for all values of the latent variable we say that the local dependence is uniform. Local dependence is often regarded as evidence of multidimensionality. It can be shown, however, that multidimensionality generates non-uniform LD. Uniform LD must therefore be interpreted as *response* dependence typically generated by the contents of items and rational response behaviour.

Graphical loglinear Rasch models: Essential validity and objectivity

There may be both several pairs of items that are locally dependent and several items that function differentially relative to one or more exogenous variables. If all DIF and LD effects are uniform they can be combined into one model referred to by Kreiner and Christensen (2002, 2004 and 2006) as a graphical loglinear Rasch model. In addition to item and person parameters, these models also have parameters describing uniform DIF and LD effects and Markov graphs where the DIF effects are indicated by edges and/or arrows between items and other variables. Figure 7 shows the IRT graph of the GLLRM fitting[5] the MH items. According to this model, there are two locally dependent sets of items, one set consisting of the two positively phrased items and another with the three negatively phrased items. “Calm” and “Downhearted” function differentially relative to Gender. The estimates of the DIF effects suggest that females are less calm, but also less downhearted than men at the same level of mental health. Three items – “Calm”, “Downhearted” and “Happy” – function differentially relative to age. The differential item functioning is first of all a question of a difference between the relatively young (18-29 years of age) and those who are 30 years or older. At the same level of mental health, the young appear to be calmer and less downhearted, but also less happy. The one exception from these trends is found among respondents of 60 years or older who appear to be even calmer than the young.

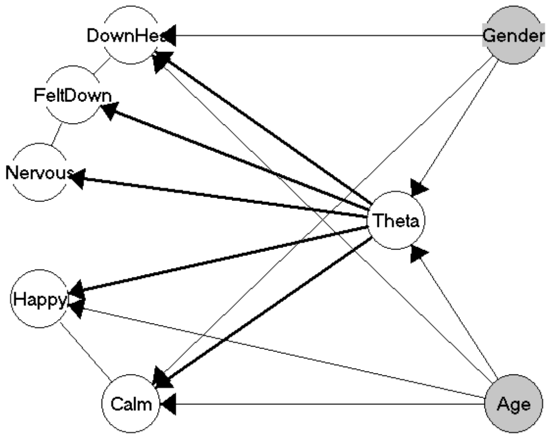

Figure 7. DIF and LD among MH items

Measurement by Graphical loglinear Rasch models satisfies unidimensionality, but the assumptions of local independence and no DIF may be violated. If some items are negatively dependent, it may even turn out that the association between items and the latent trait may not be monotonous. Measurement is therefore *not* construct valid in a formal psychometric sense. The total score of a GLLRM is, nevertheless, a sufficient statistic summarizing all information on the value of the latent variable, and sufficient statistics exist also for item difficulties, DIF parameters and LD parameters. The θ values of the latent trait variables may be estimated in principally the same way as in the conventional Rasch model, and inference on θ may be separated from inference on item, DIF and LD parameters just as inference of person and item parameters may be separated from each other in conventional Rasch models.

Kreiner and Christensen (2006) assert that measurement by items from a GLLRM is essentially valid and objective and particularly that uniform local dependence is not a problem for validity. To see this in the MH example, we define two super-items by adding locally dependent items:

Super item 1 = Calm + Happy
Super item 2 = Nervous + FeltDown + DownHearted

The result is two locally independent super-items. If there had been no DIF, the end result would have been a scale based on the two super-items satisfying all requirements of construct validity, objectivity and sufficiency. The scale would obviously have been the same as the original five-item scale that did not meet the construct validity requirements. The only way that this contradiction can be resolved in practice is by accepting that items may be uniformly locally dependent. Reliability may be affected[6], and the statistical analysis during scale validation may become more complicated, but validity is unaffected.

The DIF arguments are more difficult. The DIF relative to both Gender and Age is uniform for all affected items. This means that item responses fit a Rasch model (with or without uniform local dependence) in separate sex and age groups. Therefore, Gender and Age together define different frames of references where measurement is valid and objective within, but not across, frames of reference. It also follows that measurement would be formally valid if all the DIF affected items were eliminated. Reliability will suffer, of course, but validity will be preserved, and comparisons of persons across the different sex and age groups would be objective according to Rasch. Since the complete and reduced sets of items measure the same trait, it follows that measurements – estimates of θ values – based on the complete sets of items must also be comparable across frames of references, even though measurement is not valid.

The DIF situation is similar to the situation one has to deal with in connection with test equating between two groups of persons, where some items are shared by both groups and other items only are used in one group. Techniques for test equating are, however, readily available, and nobody argues against validity and objectivity in such situations. Test equating to adjust for DIF was described by Kreiner and Christensen (2006). Comparison of measurements is complicated, since responses to some items are missing in one group, but missing responses is only a technical problem with technical solutions and not a fundamental methodological problem of validity and objectivity. The presence of an item which is uniformly biased relative to gender, for example, presents the same technical problem as test equating, if we treat the item as two virtual items; one that was administered among men and another among women. The fact that it was the same question that was asked in the two groups has no bearing on the estimation procedure at all. It might just as well have been two different questions. Since the result of the measurement is the same, it would be inconsistent to say that measurement is valid and objective if questions are different, but not if they are the same.

We call the process of equating scores from different groups to eliminate test bias due to differential item functioning a process of DIF equating. The results will not be shown here, but as an example we can mention that DIF equating taking care of the test bias of the MH scale due to DIF relative to Gender and Age shows that at a MH score of 10 for a women older than 60 should be equated to a score of 10.9 to be comparable to the MH scores of men between 18 and 29 years of age. For a woman between 18 and 29 years the equated score is 9.9.

To distinguish between validity and objectivity as they are usually understood in psychometrics and the properties of measurement by items from graphical loglinear Rasch models, Kreiner and Christensen (2006) referred to the latter as *essential* validity and *essential* objectivity, and claimed that essential validity and objectivity were the next best to, and almost as good as, measurement satisfying all the criteria of construct validity and specific objectivity. In the best case, strict validity and objectivity obtained after elimination of items are paid for in terms of loss of reliability. In the worst case, the result may be a total loss of measurement. The fundamentalist approach to scale validation would, for instance, force us to conclude that mental health could not be measured at all using the five MH items from the SF36 questionnaire. Compared to this, measurement based on all items is perhaps not as reliable as if all items had been locally independent, but it is always more reliable than measurement based on the subset of items fitting the Rasch model. Cronbach's α, which provides a lower limit to the test-retest correlation in the study population *if items are locally independent* (Cronbach, 1951/2006*),* is equal to 0.82 for the five MH items. The test-retest correlation depends both on the distribution of the latent variables and on the item parameters[7]. It may therefore be different in different groups defined by the variables generating DIF. Examination of the test-retest correlation under the GLLRM shows that the correlation is 0.74 among the youngest, and 0.81 among the eldest. The loss of reliability due to the local dependence is apparent, but inconsequential.

Predictive validity and differential item effect

We excluded the posterior variables from the IRT graph in Figure 7 because these cannot influence measurement of mental health. The question of construct validity has therefore not been completely resolved, since the analysis based on Figure 6 could not address the question of predictive validity relative to self reported health and sleeplessness. These variables are included in the complete nomological graph, Figure 8.

The analysis of the complete analysis of the nomological network confirmed predictive validity finding highly direct effects of MH on both posterior variables. Mental health depends on Gender and Age. Consequently, both variables had indirect effects mediated by MH on both posterior variables. Age was also found to have direct effects on both variables, while Gender only had a direct effect on Sleeplessness.

During the analysis of predictive validity, one additional problem was discovered. The problem is similar to differential functioning and may be disclosed by some of the procedures that we use for DIF analyses, but the problem is better described as a problem of differential item *effect (DIE)* than as a DIF problem, because it cannot have any confounding effect on the estimation of θ values. The problem was that SRH, in addition to depending on the latent mental health variable, also depended directly on the responses to the "Happiness" item. The effect is only moderate, the partial gamma correlation between SRH and happiness controlling for Age, sleeplessness and the MH score is equal to 0.15, but the effect is nevertheless significant (p = 0.008). The DIE is that the expected self reported health of a happy person is better than the expected health of an unhappy person, even in cases where they have the same age, the same problems with sleeplessness and the same MH.

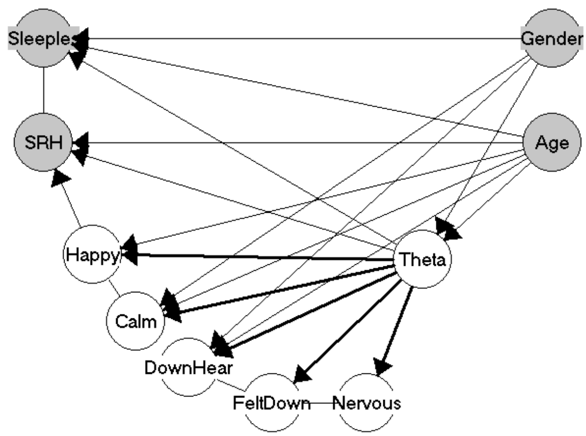

Figure 8. The nomological graph of the MH scale

DIE is fundamentally different from DIF. If it is not recognized and controlled for during an analysis of the effect of the latent variable on posterior variables, it means that the estimate of the effect will be confounded. If it is not taken care of, DIE may therefore confound the analysis of predictive validity, but it is not a measurement problem in itself. The measure, that is the estimate of the value of the latent trait, is not confounded, even though there are differential item effects.

DISCUSSION

The main points of the paper

The foremost purpose of this paper has been to discuss methodological issues rather than technical statistical problems in connection with scale development.

The paper has focused on a framework defined by a specific type of IRT models, referred to as graphical IRT and Rasch models, in which the different definitions of validity and objectivity can be treated in a unified way. The importance of understanding response behaviour has also been stressed. It is of particular importance, and often overlooked, because response behaviour as such is never included as an explicit part of the models. When evidence suggests that the fit between the data and the model is inadequate, one standard reaction is to look for and eliminate flawed items. This paper suggests that one should also consider the possibility that the problem lies with the way the model describes the response behaviour, e.g. in terms of local independence, and that items may be sound despite the lack of fit between the data and the model. Nielsen et.al. (2007) describe a scale developed partially on these principles.

The concept of differential item *effect* is new to this paper. Apart from being of some interest in itself, it was introduced to remind the reader that the question of validity of measurement is a relative question and that the question always should be rephrased as a question of whether measurement is valid with a specific purpose in mind.

Essential validity and objectivity

The existence of item response models where measurement based on sufficient scores is essentially valid and objective raises some questions concerning the appropriateness of the conventional definition of construct validity. The point of view taken in this paper is that the requirement of local independence is a requirement of convenience and not a fundamental problem for measurement. Absence of DIF is more difficult. To deal with DIF we have to assume that information on the exogenous variables with DIF effect will be available. If this is not always the case, measurement will be neither valid nor objective.

Local dependence and response behaviour

Local dependence among items is a challenge to our understanding of response behaviour of the persons when they are asked to respond to questions in a questionnaire. One conventional interpretation is that dependency always is caused by multidimensionality. Statistical tests targeting multidimensionality during the analysis of the MH scale failed, however, to identify an acceptable multidimensional structure. Notice also, that local dependencies would be non-uniform if the latent structure is multidimensional. Since there is no evidence against the hypotheses of uniform local dependencies in data, we conclude that multidimensionality does not appear to be the problem for the MH scale.

A better interpretation would be to see the local dependencies as the result of rational response behaviour aiming at logically consistent responses to the complete set of questions. The strong positive association between the responses to "Have you felt so down in the dumps that nothing could cheer you up?" and "Have you felt downhearted and low?" supports this interpretation. It is, indeed, hard to see how you could claim that you had been down in the dumps most of the time, but nevertheless had felt downhearted only a little of the time, except if you had either forgotten what you had responded to the first question or did not understand the questions at all.

The only problem with the interpretation of local dependence as a result of rational response behaviour is the lack of local dependence between these two items and the question, "Have you been a happy person". The model shown in Figure 6 is the most parsimonious model surviving the overall tests of fits of the model. We used this model for the discussion here, because it was convenient to have a model with two locally independent sets of items. Specific tests targeting local dependencies, however, disclose statistical evidence of happiness being locally dependent on both feeling downhearted and being down in the dumps.

The response behaviour must, of course, depend on the way questions are presented to respondents. SF-36 was designed for self-administration by telephone or by interview. In the current study, the questions were included as parts of a conventional questionnaire, where the respondent could see and read all questions before she answered the questions. In such situations, we can only talk about associations between responses.

One inconvenient consequence of local dependence created by rational response behaviour is that we cannot assume that the responses to the question concerning happiness would have been the same if the two other questions had not been included and that we cannot simply remove one locally dependent item from the set of items and assume that the statistical analysis of the remaining items would tell us how items would function if the excluded question had not been asked at all.

If, during an interview, the questions had been presented so that the respondent had to tell how often he had been down in the dumps before he told whether he had been downhearted and finally whether he had been happy, we must expect the response behaviour to be different in at least two different ways. The first is that responses to the first question will not be attuned to the response to the other two questions. In this case, the relationship between the three items should be regarded as a causal relationship where the response to the question concerning feeling down in the dumps has a causal effect on the response to the other two questions, and where feeling down in the dumps and being downhearted have an effect on the response to the question concerning happiness. If data had been collected in this way, an analysis without the two subsequent questions would have told us how the first of the three questions would have functioned in situations where the two other questions had not been asked at all. The second is that the strength of the local dependency between item responses collected during an interview cannot be expected to be the same, because the respondent may be unable to completely recall both the phrasing of the first question and precisely which of the six response categories that were selected. We cannot, in other words, assume that MH scores calculated from data collected during the survey providing the data for the example discussed in this paper are comparable to MH scores collected during interviews of persons from the same population.

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NOTES

1 Journal of Applied Measurement and Journal of Outcome Measurement both published by JAM press

2 Personal communication, first from Benny Karpatchof who introduced me to Rasch models in 1969 and later by Rasch himself.

3 Also known as modern test theory.

4 For the psychometrically inclined reader, we have included some results from the analysis of the MH scale in Appendix C.

5 The tests of fit supporting this model are included in Appendix C.

6 Reliability measured as the test-retest correlation decreases while Cronbach's α increases, if items are positively locally dependent. Imagine an extreme case where the same question is asked five times. The reliability of the total score on these items is not better than the reliability of one question, but Cronbach's α is equal to 1. In other words, Cronbach's α is no longer a lower level of the true reliability if items are positively locally dependent.

7 The distribution of the score depends on the item parameters, from which it follows that the distribution of repeated measurements of the score, and therefore the correlation, also depends on the item parameters.

Appendix A. Acronyms

The acronyms used throughout this paper are shown below

Acronym	Meaning
CLR	Conditional likelihood ratio test
DIE	Differential item effect
DIF	Differential item functioning
GLLRM	Graphical loglinear Rasch model
LD	Local dependence
LI	Local independence
MH	Mental health
PCM	Partial credit model
SF-36	Short form 36
SRH	Self reported health

Appendix B. The Rasch model

The Rasch model for dichotomous items is usually parameterized as shown in formula (B.1)

$$P(Y_i = 1 \mid \theta) = \frac{\exp(\theta - \alpha_i)}{1 + \exp(\theta - \alpha_i)} \qquad \text{(B.1)}$$

The model is the same as the same model shown in Figure 1 or as formula (1) with the common item discrimination parameter set to zero. The α-parameters in (B.1) are usually called item thresholds and regarded as measures of item difficulties on the same interval scale as the latent trait variable. The threshold of an item is equal to the value of the latent trait where the probability of a positive response is equal to 0.5.

Unidimensional Rasch models for polytomous ordinal items are often called partial credit models (PCM). The probabilities of responses in the PCM model are defined in formula (B.2) where we assume that the k item response categories are integer coded from 0 to k-1.

$$P(Y_i = y \mid \theta) = \frac{\exp\left(\sum_{x=0}^{y} (\theta - \alpha_{ix})\right)}{\sum_{z=0}^{k-1}\left(\exp\left(\sum_{x=0}^{z} (\theta - \alpha_{ix})\right)\right)} \qquad \text{(B.2)}$$

The PCM model is obviously more complicated than the Rasch model for dichotomous items. The symmetry of items and persons in the Rasch model for dichotomous items, which was so important to Rasch (1960/80) when he introduced the model with one person and one item parameter and the interpretation of item parameters in terms of item difficulties, break down. The item parameters, $\alpha_{i0}, \ldots, \alpha_{ik-1}$ are also referred to as item thresholds posited on the same interval scale as the values of the latent trait, and the interpretation of the meaning of the item thresholds is somewhat controversial.

The PCM model has the same fundamental measurement properties as the Rasch model for dichotomous items. The total score is a sufficient statistic for the latent trait, and inference on the values and distribution of θ may be separated from inference on the item parameters.

Uniform DIF

An item functions differentially if the item in addition to depending on the latent variable also depends on an exogenous variable, X, referred to as the DIF source. If the size of the effect of the DIF source on the item does not depend on the latent variable we say that the DIF is uniform. The probability of a positive response on a dichotomous item may in this case may be written as

$$P(Y_i = 1 \mid \theta, X = x) = \frac{\exp(\alpha_i + \beta_i\theta + \beta_X x)}{1 + \exp(\alpha_i + \beta_i\theta + \beta_X x)} \qquad \text{(B3)}$$

Where α_i and β_i in formula (3) are item parameters as in the conventional Rasch model, describing the difficulty and discrimination of the item and β_X is the effect of X on the item.

Uniform LD

Kreiner and Christensen (2002) define uniform local dependence in a similar way by adding interactions between item responses that do not depend on the value of the latent variable to the model. The joint probability of positive responses to items i and j is shown in formula (4) where β_{ij} is the parameter describing the uniform local dependence between the two items. The formula is reduced to the product of probabilities for positive two responses if $\beta_{ij} = 0$.

$$P(Y_i = 1, Y_j = 1 \mid \theta) = \frac{\exp(\alpha_i + \alpha_j + (\beta_i + \beta_j)\theta + \beta_j)}{1 + \exp(\alpha_i + \beta_i\theta) + \exp(\alpha_j + \beta_j\theta) + \exp(\alpha_i + \alpha_j + (\beta_i + \beta_j)\theta + \beta_j)} \qquad \text{(B4)}$$

Graphical loglinear Rasch models

(B3) and (B4) may be generalized to models where several items function uniformly differentially relative to several different DIF sources and where several pairs of variables are uniformly locally dependent. These models are referred to as graphical loglinear Rasch models because the joint distribution of item responses given the latent variable and given the exogenous variables is similar to a loglinear model for a multidimensional contingency table.

Appendix C. Item analysis of the MH scale

The program DIGRAM (Kreiner, 2003) was used for the analysis of the MH scale. This appendix shows some, but far from all, evidence against the conventional graphical Rasch model (GRM) and for the graphical loglinear Rasch model (GLLRM) shown in Figure 6. In order to take multiple testing into account, the significance of fit statistics was assessed by the Benjamini-Hochberg (1995) procedure with a false discovery rate at 0.05

The conditional likelihood ratio test of Andersen (1973) is an overall test-of-fit comparing estimates of item parameters in different groups. CLR tests are used both for analysis of item homogeneity across score groups and for overall test of no DIF by comparison of parameters in groups defined by exogenous variables. The test results in Table C.1 relating to the graphical Rasch show that item responses are heterogeneous across score groups and DIF affected relative to Age. The CLR test of the graphical loglinear Rasch model shown in Figure 6 is, however, accepted.

Table C.1 *Conditional likelihood ratio (CLR) tests comparing estimates of item parameters in different groups.*

	GRM			GLLRM		
Groups	CLR	df	p	CLR	df	p
Score groups 1-20 vs. 21-24	207.0	24	<0.0005	128.4	151	.908
Gender	34.9	24	0.070	158.7	131	0.050
Age	305.7	72	<0.0005	318.1	273	0.031

Many item-fit statistics have been proposed. One of the most powerful is the correlation between item responses and rest scores – the score on all other items. Table C.2 shows both the observed correlations and the expected correlation under both the Rasch model and the graphical loglinear Rasch model. Comparison between observed and expected correlations discloses evidence against the GRM, but not against the GLLRM.

Table C.2 *Observed and expected gamma correlation coefficients measuring association between item responses and rest scores. Numbers in parentheses after expected values are p-values from tests comparing observed to expected values.*

Item	Observed	Expected under GRM	Expected under GLLRM
NERVOUS	0.598	0.633 (0.0213)	0.593 (0.7338)
FELTDOWN	0.770	0.662 (<0.00005)	0.765 (0.7387)
CALM	0.649	0.663 (0.2129)	0.673 (0.0549)
DOWNHEAR	0.661	0.632 (0.0333)	0.629 (0.0239)
HAPPY	0.677	0.661 (0.1931)	0.668 (0.5107)

The evidence of local dependence and differential item functioning is overwhelming. Table C.3 shows conditional likelihood ratio test for each of the LD and DIF effects in the model. Additional information on these tests may be found in Kreiner and Christensen (2002, 2004,2006)

Table C.3 *Tests of local independence and no DIF under the GLLRM shown in Figure 6.*

Effect	Variables	CLR	df	p
Local dependence	NERVOUS and FELTDOWN	90.0	25	<0.00005
	FELTDOWN and DOWNHEART	251.0	25	<0.00005
	CALM and HAPPY	395.3	25	<0.00005
DIF	CALM and Gender	19.0	2	0.0019
	CALM and Age	86.9	15	<0.00005
	DOWNHEART and Gender	19.5	5	0.0016
	DOWNHEART and Age	80.9	15	<0.00005
	HAPPY and Age	62.2	15	<0.00005